

Muhammad Tayyeb Bukhari

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EDUCATION

Lahore University of Management Sciences (LUMS)

BS Computer Science — Sophomore, GPA: 4.0 / 4.0

2025 – Present

Lahore, Pakistan

Beaconhouse College Programme, Abbottabad

Cambridge A-Levels — Rank 2 / 160

2023 – 2024

- Physics A*, Chemistry A*, Biology A*, Mathematics A

EXPERIENCE

SPADES (LUMS Student Society) — AI/ML & Robotics

2025 – Present

Project Lead, Robotic Arm Imitation Learning; Team Member, AI/ML Department

- Leading a project on robotic arm imitation learning, developing control pipelines for physical manipulation tasks
- Contributed to AI/ML department initiatives and collaborative research within the society

LUMS — NERC (National Engineering Robotics Contest)

2025 – Present

Robotics Team Member

- Designed, built, and programmed an NERC competition robot; secured funding from the LUMS EE Department

Xylexa.ai

2024

AI/ML Intern

- Developed AI algorithms for detecting 14 chest CT abnormalities; addressed dataset imbalance via oversampling and augmentation
- Enhanced breast cancer detection pipelines and built triaging systems for urgent care prioritization

National Centre of Robotics and Automation (NCRA)

2024

AI Research Intern

- Built AI models for medical datasets: lumbar degeneration classification, chest X-ray anomaly detection, seizure detection from EEG
- Applied GANs and advanced vision techniques; authored and published research on medRxiv

Combined Military Hospital (CMH) — AI Tumor Detection Project

2023

Lead Developer & Researcher

- Designed a SqueezeNet-based neural network achieving 97% accuracy in tumor detection; received CMH certification
- Produced diagnostic reports and visual aids for clinical staff

PROJECTS

AI Agentic Chatbot

github.com/Yeetsnake2/AI-agentic-chatbot

- Built a Gemini-based chatbot with web search integration and semantic document parsing in Python

Pneumonia Diagnosis Classification

github.com/Yeetsnake2/Pneumonia-diagnosis-classification-model

- Compared four lightweight models (EfficientNet-Lite0, MobileNetV3, MobileViT, EfficientFormerV2) for pneumonia detection; MobileViT achieved 97.5% accuracy

Tokamak Simulation

github.com/Yeetsnake2/Tokamak-simulation

- OpenGL compute-accelerated plasma simulation of a tokamak fusion reactor

Rubik's Cube Solver Robot

2023

- 3D-printed, soldered, and programmed a fully autonomous cube solver for under \$90; achieved 95% solving efficiency; covered by 2 news outlets

PUBLICATION

Comparison of Lightweight CNN and Hybrid ViT Models for Pneumonia Detection

medRxiv, Oct 2024 · [doi:10.1101/2024.10.24.24316057](https://doi.org/10.1101/2024.10.24.24316057)

LEADERSHIP & ACTIVITIES

LSMS (LUMS Math Society) — Event Head

2025 – Present

Sigma and Delta Olympiads — two of Pakistan's largest math olympiads

- Epilepsy Awareness KPK — Founder & Project Lead**

2024 – Present

 - Raised epilepsy awareness across KPK; collaborated with hospitals and raised funds; reached 20K+ people/month
- Robotics & CS Society, Beaconhouse — President**

2023 – 2024

 - Led 60+ member society; taught Python, C++, CNNs, 3D modeling, and circuit design; organized COMSATS IoT bootcamp
- National Science Talent Contest — Alumni Mentor**

2024 – Present

 - Coached IPHO and INSO aspirants; compiled resource banks, designed problem sets, and held training sessions

HONORS & AWARDS

- Gold Medal — International Nuclear Science Olympiad (IAEA) · First Gold for Pakistan

2025
- Pride of Pakistan — conferred by the Government of Pakistan

2024
- Top 5 Nationally — National Physics Talent Contest, HEC Pakistan (ranked 5th)

2024
- Silver Medal — Copernicus Physics & Astronomy Olympiad

2024
- Top 10 — National Science Olympiad III (Great Future Pakistan)

2024

SKILLS

- Languages:** Python, C++, C, JavaScript
- ML / AI:** PyTorch, TensorFlow, scikit-learn, OpenCV, CNNs, ViTs, GANs, imitation learning
- Tools:** Git, Linux, OpenGL, Jupyter, Alumentations, FLTK
- Other:** 3D prototyping, circuit design (Fritzing), embedded systems, robotics